

Harnessing Biomechanical Energy Using Shoulder Exoskeleton: Electrical Energy Regeneration

Ali Nasr¹

Postdoctoral Research Associate,
Systems Design Engineering
Department,
University of Waterloo,
Waterloo, ON, Canada.
email: a.nasr@uwaterloo.ca

John McPhee

ASME Fellow,
Professor and Canada Research
Chair,
Systems Design Engineering
Department,
University of Waterloo,
Waterloo, ON, Canada.
email: macphee@uwaterloo.ca

Implementing back-drivable actuators with energy regeneration capabilities can improve the efficiency and extend the battery life of robotic exoskeletons, by converting a portion of dissipated energy during negative mechanical work into electrical energy. We present the first study of the regeneration capabilities of shoulder exoskeletons, which includes experimental motor parameter identification, the development of an electro-mechanical upper-limb human-exoskeleton model for energy regeneration, and task-based experiments assessing real-world energy generation despite inherent motor limitations. Initial steps involve motor parameters identification using a joint dynamometer system and modelling of the human-exoskeleton system with a specific emphasis on energy regeneration in the shoulder region. Task-based experiments were conducted to evaluate energy regeneration during daily exoskeleton usage. Our experimental identification reveals that the regenerated current follows a third-degree polynomial relationship with joint angular speed. Task-based simulations indicate that electrical energy recovery ranges from 30.4 μJ during slow motions to 1.02 mJ during very fast movements. Efficiency analysis shows that performance peaks at approximately $60^\circ/\text{s}$, where the trade-off between increased current generation and rising frictional losses is optimized. Despite challenges such as limited electricity generation due to motor characteristics, our findings demonstrate the feasibility of energy regeneration in shoulder exoskeletons, albeit with certain constraints.

Keywords: Energy harvesting, Energy regeneration, Wearable robotics, Exoskeletons

1 Introduction

Upper-limb exoskeletons have valuable applications in rehabilitation settings [1], assisting individuals with object manipulations [2], strength recovery by providing targeted support, and enhancing the effectiveness of physical therapy exercises [3]. In industrial or military applications, the portability of these robots becomes important, enabling fast and easy deployment and adaptability in dynamic environments [4]. This enhances operational flexibility, efficiency, and responsiveness to changing demands or scenarios.

Exoskeletons draw power from various sources, including passive [5], active [6,7], and active-passive [8] configurations. Occasionally, these systems are engineered to operate self-sufficiently independent of external electrical power consumption [9]. Note that while passive exoskeletons contribute to the array of options, they are limited to providing a fixed torque profile [5]. In contrast, the majority of active exoskeletons utilize electromagnetic actuators, specifically employing brushed or brushless direct current (BLDC) motors [8,10]. These motors are often paired with a high-ratio transmission, such as harmonic gearing or a ball-screw mechanism, to amplify the motor torque output essential for human movement. Consequently, the robotic actuator exhibits heightened output impedance [11,12].

Conventional rigid actuators are constrained in their ability to exploit the inherent passive dynamics of human motion or integrate other energy storage and return mechanisms. This limitation often results in the creation of bulky and inefficient actuators, demanding substantial energy consumption and thereby restricting the operational duration of battery-powered systems due to the finite energy density of rechargeable batteries [11]. The majority of robotic exoskeletons typically provide a maximum battery-powered runtime ranging from 1 to 5 hours [3]. The development of robotic exoskeletons for practical mobility has consistently identified on-board portable power as a primary impediment [3]. The increased mass and inertia of the device, associated with heavier onboard mo-

tors and batteries, necessitates heightened exertion from the human musculoskeletal (MSK) system [11]. Consequently, this diminishes human motion efficiency by leading to higher metabolic power consumption [11]. To address these challenges, scientists are actively involved in the development of lightweight and efficient actuators aimed at better harnessing the energy dynamics inherent in human motion.

Exoskeleton robots commonly employ torque-dense motors with low-ratio transmissions, known as quasi-direct drives, to achieve desirable characteristics such as low mechanical impedance, high backdrivability, and optimal efficiency [13,14]. In the drone industry, the preference for reduced transmission ratios predominantly arises from recent advancements in high-torque motors. These actuators generate substantial output torque by enhancing motor torque density rather than adjusting the transmission ratio, effectively mitigating undesirable outcomes associated with high gearing, such as increased weight, backlash, and reflected inertia, which scales proportionally with the square of the transmission ratio [12]. Gears also make the control more challenging, introduce torque-dependent friction, contributing to heightened impedance and reduced backdrivability and efficiency [15]. Overcoming actuator backdriving resistance necessitates significant external forces. The characteristics of substantial gearing may impede dynamic physical interactions, both between the human and the device and between the device and its surroundings [8]. This limitation presents challenges, particularly for individuals with partial motor control function, such as older adults and/or those with conditions like osteoarthritis or post-stroke, who could benefit from the active backdriving capability of the joints [11].

Actuators characterized by both backdrivability and low output impedance offer several advantages for control and efficiency, encompassing: 1) unrestricted dynamic limb motion, facilitating simplified control during the free motion and promoting energy-efficient movement; 2) compliance in handling impacts; 3) minimal actuator dynamics, contributing to simplified control; 4) inherent backdrivability; and 5) the potential for energy regeneration during periods of negative mechanical work [13]. Energy regeneration in-

¹Corresponding Author.

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volves transforming a portion of the energy typically dissipated during negative mechanical work into electrical energy by backdriving the actuator. Put differently, when subjected to external backdriving forces, the motor can exert a braking torque to slow down the load (such as motion control during the swing phase) and simultaneously produce electricity [11]. This is comparable to the regenerative braking systems employed in electric and hybrid electric vehicles. Given a suitable motor driver, the recovered energy has the potential to recharge batteries support power generation in other joints. These design principles were applied in the MIT Cheetah robot [13] and, more recently, in lower wearable robotic devices [8,14,16–19].

While regenerative actuators offer potential benefits in enhancing exoskeleton efficiency, prolonging battery-powered operational duration and reducing the weight of onboard batteries, existing research has focused on scenarios involving lower-limb exoskeletons during walking, mostly for the knee joint [7,9,14,17,20–28]. Specifically, in the context of ambulation, the human body engages in both positive and negative work during the phases of progression and deceleration, respectively [23–26]. Consequently, the effective utilization of negative work incurred during deceleration, through the incorporation of auxiliary devices designed to augment the subsequent acceleration of lower limb movement, holds the potential to mitigate the overall metabolic expenditure associated with human ambulation [2,9,29]. Laschowski et al [11] conducted an evaluation of negative work and assumed that all energy would be stored electrically using a non-backdrivable exoskeleton, which is not practical.

A generator integrated into an exoskeleton, functioning as a damping element, can capture kinetic energy to generate electrical power for the operation of wearable electronic devices [7,9,14,21,24–26]. Sometimes, these systems utilize a torsion spring and clutches to regulate mechanical energy recycling in a gait cycle [23]. To our knowledge, previous studies have not investigated electricity regeneration in upper-limb exoskeletons, underscoring the unique focus of our work.

Real-world movement involves a diverse range of activities, including object manipulation at varying speeds, environment interaction, and seamless transitions between different modes of motion [30]. Dynamic situations, such as shifts between task modes and variations in speeds and external object weight, play a crucial role in modeling human motion and developing upper-limb wearable robotic devices. Motivated to explore energy regeneration in various daily motion tasks, we have developed an electro-mechanical model for an energy regeneration system involving a human-exoskeleton. This model is specifically designed to simulate the process of backdriving a shoulder exoskeleton using experimental data from joints of the upper-limb. To validate the performance, assessments of the shoulder exoskeleton device were conducted using a joint dynamometer system. This system, in conjunction with an electro-mechanical motor model, facilitated the computation of actuator efficiency and the simulation of energy regeneration and storage during human upper-body movement.

While numerous studies have demonstrated the feasibility of energy regeneration in lower-limb exoskeletons during walking—most notably through biomechanical energy harvesting techniques [10,11,19,20,23,31,32]—the application of similar regenerative strategies to upper-limb devices remains largely unexplored. The shoulder joint exhibits complex, multi-axis motion and a broader range of dynamic movements during tasks such as object manipulation and overhead work, presenting unique challenges compared to the relatively linear kinematics of the knee [33]. Our study addresses this gap by developing a dedicated biomechatronic model (for simulation) that accounts for the intricacies of human-exoskeleton dynamics, as well as experimental identification of an actual motor.

In summary, this research has the following contributions:

- Developed a multibody dynamic model specifically tailored for an energy regeneration system, encompassing human (upper-limb)-(shoulder) exoskeleton sub-systems;
- Conducted identification and performance assessments of the

shoulder exoskeleton device utilizing a joint dynamometer system;

- Evaluated object manipulation and over-head work experimental results against realistic scenarios, providing a robust assessment of energy regeneration performance.

In this study, section 2 delineates the multibody upper-limb human-exoskeleton system, with a specific emphasis on the BLDC motor. Following this, in section 3, we present two experimental setups designed for the motor identification and upper-body motion capture. Lastly, section 4 presents the results of our experiments, accompanied by a discussion that highlights additional challenges necessitating attention in future studies.

2 Electro-mechanical model

2.1 Multibody human-exoskeleton dynamic model. The modeled human-exoskeleton system comprises two main components: the human body and a wearable robot.

A 20-degree of freedom (DoF) scalable MSK upper body model was created, comprising an 8-segment model validated using experimental dynamometer data. The MapleSim model includes a skeletal dynamic model and biomechanical joint torque models based on muscle torque generators, incorporating active units for torque generation in multiple-DoFs or anatomical directions [34,35].

The wearable robot has 14 passive revolute joints and is powered by a 9 N m BLDC motor. The closed kinematic loop of the system involves connections between the exoskeleton and the human body at the belt and armrest, resulting in a set of scalar constraint equations that enforce the human-exoskeleton coupling [8].

Equations (1–4) represent the system dynamics as a system of differential-algebraic equations, considering the dependence of generalized coordinates (q) and generalized speeds (p). A visual representation of the human-exoskeleton system is shown in Figure 1(c).

$$M\dot{p} + C^T\lambda = F + Q \quad (1)$$

$$\Phi(q, t) = 0 \quad (2)$$

$$\Psi(p, q, t) = 0 \quad (3)$$

$$\chi(\dot{p}, p, q, t) = 0 \quad (4)$$

where

- n number of independent coordinates = 30 (20 human joint angles and 10 robot joint states)
- m number of constraint reactions = 10 (5 for each side)
- M $n \times n$ mass matrix, a combination of the human model, the exoskeleton, and the mass matrices of the assistive components
- C $m \times n$ constraint Jacobian (maps reaction forces/torques to generalized forces)
- λ $m \times 1$ reaction forces and torques in the co-tree joints that enforce the kinematic constraint of human-exoskeletons connection
- F the $n \times 1$ right-hand side of dynamic equations, not including the generalized force vector
- Q the $n \times 1$ generalized force vector containing the applied wrench (torque/force) at the joints, which consists of the human joint torque (τ) and the robot joint torque (T)
- Φ $m \times 1$ position-level kinematic constraint equations
- Ψ $m \times 1$ velocity-level kinematic constraint equations
- χ $m \times 1$ acceleration-level kinematic constraint equations
- q n generalized coordinates of the multibody system, which consists of the human joint angles (θ) and the robot joint states (φ)
- p n generalized speeds of the multibody system, which consists of the human joint angular velocities (ω) and the derivative of robot joint states ($\dot{\varphi}$)

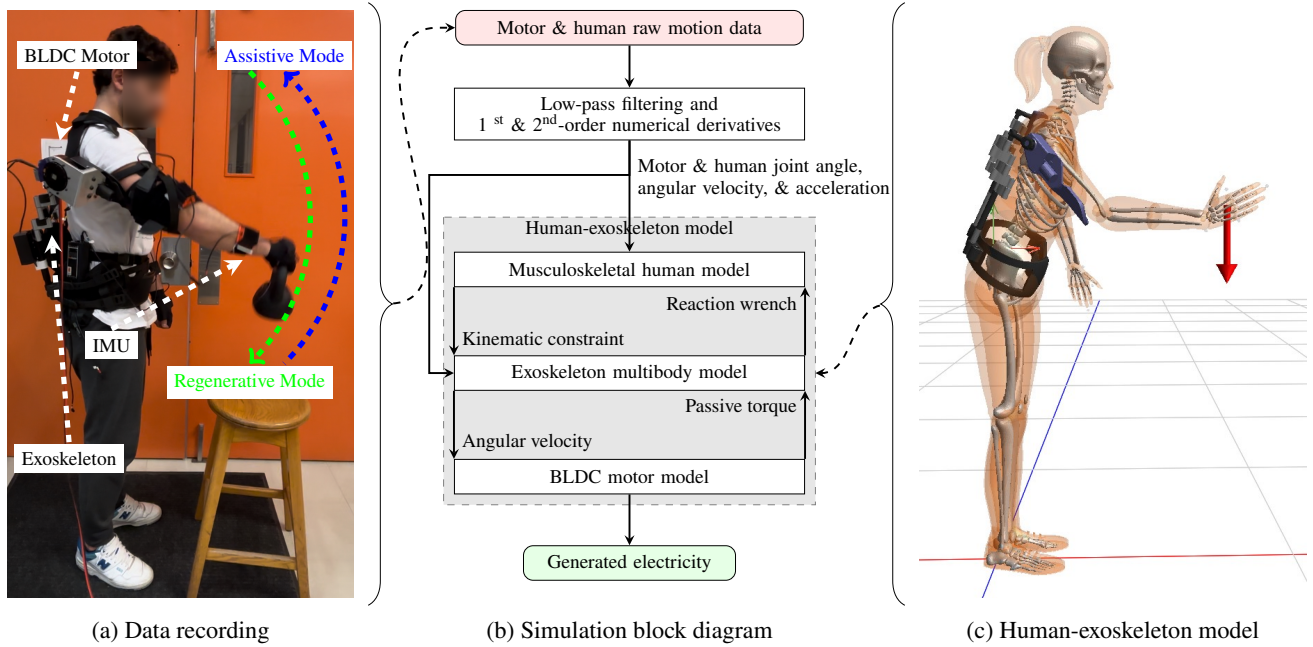


Fig. 1 (a) Motion data recording during overhead work by a healthy subject (exoskeleton assistive mode inactive), (b) block diagram of inverse dynamic simulation workflow, and (c) scalable upper body MSK model integrated with exoskeleton dynamics.

2.2 BLDC motor model. A BLDC motor, also known as a permanent magnet synchronous motor, operates differently from a traditional brushed DC motor. However, the principles of operation as a generator in passive mode are similar, with some variations in the mathematical model due to the motor's construction. A BLDC motor can indeed act as a generator when it's being mechanically driven. This is often referred to as regenerative braking or passive mode operation [29]. The mathematical model for a BLDC motor operating in this mode involves several parameters and equations.

The voltage (V_{emf}) generated by the BLDC motor acting as a generator is given by the back electromotive force (EMF), which is proportional to the angular velocity (ω) of the motor. This can be represented as Equation (5). Equation (6) shows that the generated current (I) is determined by the difference between the back EMF and the measured voltage (V_m), divided by the load resistance (R). This implies that, as the angular velocity or back EMF constant increases, the generated current will also increase, provided that the load resistance and the measured voltage remain constant. However, practical considerations such as the motor's internal resistance, the type of load, control mechanisms, and nonlinearities such as magnetic saturation or control loop delays, can influence the actual relationship between angular velocity and generated current, leading us to replace Equation (6) with a nonlinear (polynomial) model as presented in Equation (7) for this study.

The torque produced by the generator (T) is proportional to the current (I) flowing through it, and this current is related to the load connected to the generator. The regenerative torque is given by Equation (8), where k_t converts the current into torque.

The electrical power (P_e) generated by the motor can be calculated using the voltage and current as Equation (9). While, the mechanical power (P_m) input to the generator is related to the torque and angular velocity as described in Equation (10). In passive mode, where the motor is generating electricity while being driven by an external force, the mechanical power input (P_m) is negative because the motor is acting as a load. Therefore, the electrical power generated (P_e) is positive, indicating power being delivered to the load connected to the generator.

The efficiency of the generator system can be calculated as the ratio of electrical power output to mechanical power input as Equa-

tion (11). This efficiency factor accounts for losses in the system, including mechanical losses, electrical losses, and any losses due to the conversion process. Keep in mind that the actual performance of a DC motor operating in passive mode depends on various factors such as the motor's characteristics, the load connected to it, and the speed at which it's being driven. These equations provide a basic framework for understanding the electrical generation capability of a BLDC motor in passive mode.

$$V_{emf} = k_e \omega \quad (5)$$

$$I = \frac{V_{emf} - V_m}{R} = \frac{k_e \omega - V_m}{R} \quad (6)$$

$$I = \sum_{i=1}^n p_i \omega^i \quad (7)$$

$$T = k_t I = k_t \frac{(V_{emf} - V_m)}{R} = k_t \frac{(k_e \omega - V_m)}{R} \quad (8)$$

$$P_e = V_m I \quad (9)$$

$$P_m = T \omega \quad (10)$$

$$\eta = -\frac{P_e}{P_m} = -\frac{V_m I}{T \omega} \quad (11)$$

$$E = \int_{t_{start}}^{t_{final}} P_e dt \quad (12)$$

where

V_{emf} the ideal voltage generated by the motor's rotation (V)
 V_m the voltage measured across the resistor (or load) (V)
 R the load resistance (Ω)
 ω the motor's angular velocity (rad/s)
 k_e the motor's back EMF constant (Vs/rad)
 k_t the motor's torque constant (Nm/A)
 p_i the term of coefficients

n	the maximum number of coefficients
T	the torque generated (N m)
P_e	the electrical power generated by the motor (W)
P_m	the mechanical power input (W)
η	the efficiency of the generator system
E	the generated energy (J)

3 Experiments and simulation evaluation

This section presents the experimental and simulation procedures conducted to evaluate electrical energy regeneration performance of the human-exoskeleton system. The evaluation was divided into two phases: (1) The first phase (section 3.2), termed identification measurement, aimed to characterize the current-motion-torque interdependence of the BLDC motor, and (2) the task-based experiment phase (section 3.3), which involved human-participant testing to assess the exoskeleton's functionality.

A variety of equipment was employed to capture the necessary mechanical, electrical, and kinematic data. The identification measurement phase utilized a joint dynamometer to control and measure joint torque and angular velocity, along with a digital multimeter to quantify electrical current generation. For the task-based experiment, a wearable inertial measurement unit (IMU) motion capture system recorded joint kinematics, while onboard sensors of the BLDC motor provided complementary angle measurements. All equipment was calibrated and synchronized to ensure high-fidelity data acquisition. The subsequent subsections detail the equipment specifications, experimental protocols, and data processing methods.

3.1 Equipment. The equipment utilized in the experimental test are introduced as follows:

3.1.1 Shoulder exoskeleton. In this investigation, we employed an upper-limb exoskeleton, specifically the EVO model (Ekso Bionics Holdings Inc, California, USA) as the wearable device equipped with passive assistance that can be deactivated as needed. Reverse engineering techniques were employed to create computer-aided geometric models, determining precise dimensions, joint locations, mass, center of mass, and the inertia matrix. The focus was on the exoskeleton's elevation mechanism and assistive torque, with detailed information available in [8].

3.1.2 BLDC motor. The motor was an AK80-9 KV100 BLDC motor (Cubemars, Jiangxi Xintuo Enterprise Co., China) equipped with a built-in relative encoder. It has a mass of 0.485 kg, a rated torque of 9 N m, and a gear ratio of 9 : 1 for either assistance or electricity generation. Control of the active assistance (motor) is by a hierarchical control structure utilizing a subject-specific surface electromyography-driven intention prediction model [36,37]. Note that the commanded torque (the desired output torque from the motor to the exoskeleton joint) should always be less than the peak torque of 18 N m, which represents the maximum torque the motor can produce briefly, usually during acceleration, deceleration, or when overcoming friction. Additionally, the achievable speed decreases as the commanded torque increases, and the motor temperature rises.

3.1.3 Dynamometer. We employed a joint isokinetic dynamometer (Biodex Medical Systems, Inc, Shirley, NY) for assessing the passive resistive torque of the motor and exoskeleton. These dynamometers, commonly utilized for isolated, single-joint assessments, enable the evaluation of passive, torque-angle, and torque-angular velocity relationships [38]. These relationships are important for calculating joint mechanical friction, work, and power. Specifically, the exoskeleton elevation joint was tested, with straps employed to secure the exoskeleton fixture to the dynamometer attachment. This limitation relative movement and misalignment between the rotational axes of the exoskeleton

and the dynamometer shaft, which could lead to errors in torque measurements [38].

3.1.4 Digital multimeter. In instances where the motor current was exceptionally low, a resistor was employed within the electrical circuits. The integration of a resistor enabled the transformation of the minute current into a more easily detectable voltage, thereby facilitating a more precise and reliable measurement. The voltage was measured using a Keysight/Agilent 34411A digital multimeter (Keysight Technologies, Santa Rosa, CA, USA), which has a basic accuracy of 0.0030 % DC and 50,000 readings/sec.

3.1.5 Motion capture. Angular joint kinematics during the movement were recorded using an IMU system (MVN, Xsens Technologies, Netherlands). Comprising 11 IMUs for upper-body or 17 IMUs for full-body, these were affixed to various body segments, including the head, torso, upper arms, forearms, hands, thighs, shanks, and feet. Each IMU incorporates a triaxial accelerometer, rate gyroscope, and magnetometer [39]. The accelerometers measure accelerations, including gravitational acceleration, the magnetometers measure the geomagnetic field, while the rate gyroscopes capture angular velocities. Employing a 23-segment biomechanical model and proprietary algorithms, the IMU system calculates angular joint kinematics [39].

3.2 Identification measurement. To identify the current-motion-torque interdependence of the BLDC motor, an experimental identification test is systematically executed, as delineated in Figure 2. The obtained data support subsequent analyses and optimizations to enhance the overall efficiency and sustainability of the BLDC motor system.

Prior to conducting the identification tests, the dynamometer was calibrated according to the manufacturer's guidelines to ensure accurate torque and angular velocity measurements. The BLDC motor was aligned with the dynamometer shaft using adjustable fixtures. To achieve precise alignment, a digital level and an angle meter were employed to verify both angular and positional consistency between the motor and dynamometer shafts. The alignment was double-checked at multiple stages to ensure that even minor deviations were corrected, thereby reducing the risk of measurement errors caused by shaft misalignment during the tests. Environmental conditions were controlled, with room temperature maintained at approximately 19 °C to minimize variations in material properties and sensor performance.

The digital multimeter was set to a sampling rate of 50,000 readings per second. Its connection to the 100 Ω resistor (rated at 0.5 W and confirmed via brown-black-brown-gold coding) was verified before each trial to ensure accurate current measurements. Synchronization between the dynamometer and multimeter was accomplished through convolutional techniques, ensuring proper alignment of dynamic signals.

The testing procedure involves the deactivation of the motor, following which a dynamometer imparts controlled motion to the motor at a specified velocity. Concurrently, the resulting passive torque (T_p) is measured, providing insights into the motor's dynamic behavior. Simultaneously, during this controlled motion, a digital multimeter is employed to measure and quantify the electrical energy generated by the motor. This measured electricity is directed and stored in the battery, an important step in understanding the efficiency and performance characteristics of the BLDC motor. Note that a similar experiment was conducted in [11] without measuring electrical energy generation.

The exoskeleton elevation joint underwent 10 sinusoidal flexion and extension movements using position control with the dynamometer. The experimental procedure was repeated 10 times with angular velocity of 10, 20, 30, 45, 60, 75, 90, 120, 150, 180 °/s, with 5-minute intervals between sessions to prevent overheating of the exoskeleton motor. The isokinetic mode of the dynamometer was selected to drive the exoskeleton motor shaft throughout the range of motion without interference.

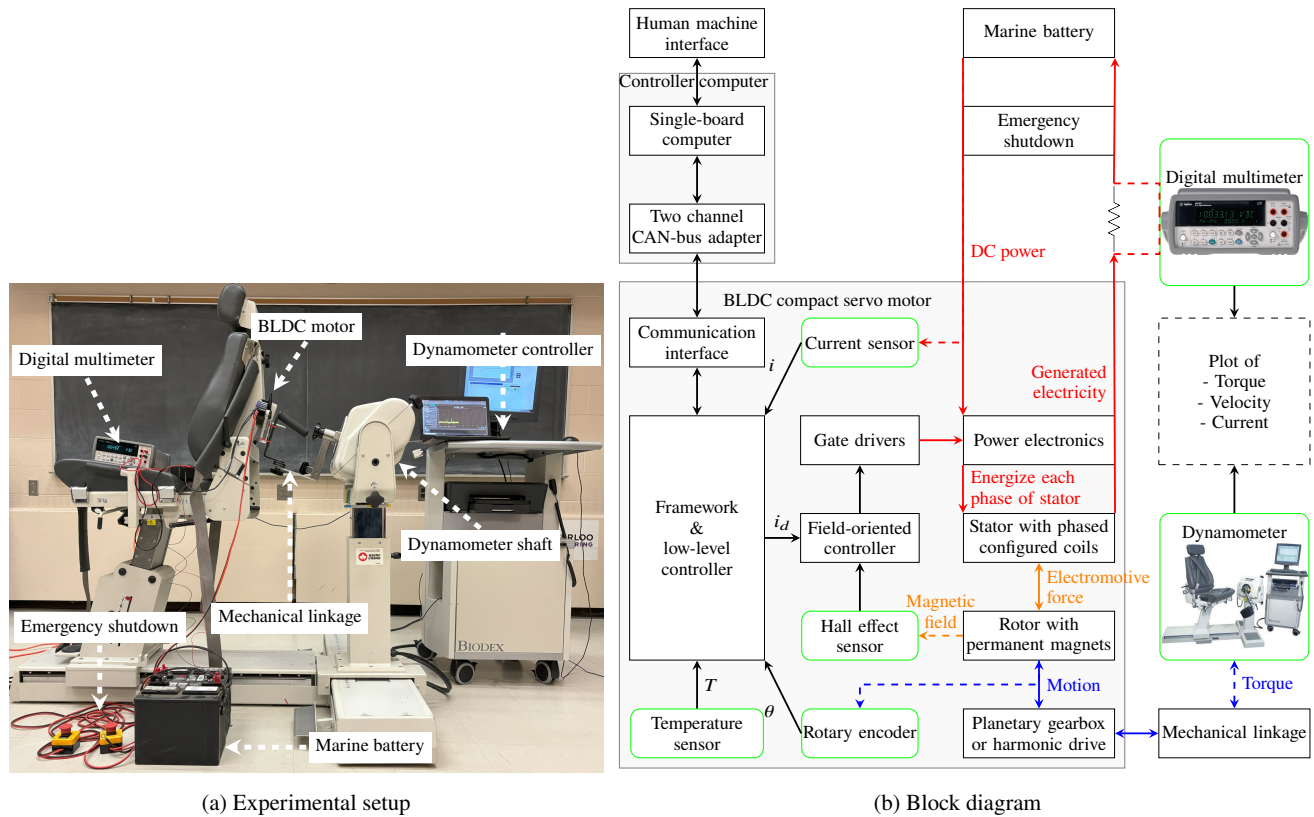


Fig. 2 Energy generation of BLDC motor testing: utilizing a joint dynamometer robotic system and a digital multimeter. The exoskeleton structure undergoes sinusoidal flexion and extension motions, with the dynamometer measuring joint resistive torque and angular speed. The digital multimeter measures generated electricity current indirectly from the voltage across the resistor. (a) Experimental setup (left) and (b) block diagram (right). Data representations: sensor/measurement units (green rounded boxes), other components/units (black solid rectangles), offline processes (dashed rectangles), electrical relationships (red arrows/lines), magnetic relationships (orange arrows/lines), force/torque/motion relationships (blue arrows/lines), and data relationships (black arrows/lines).

3.3 Task-based experiment. Experimental motion data obtained from a subject was utilized in the inverse dynamic simulation of a human-exoskeleton multibody dynamic model, as described below.

A healthy individual (male, aged 25, 1.7 m tall, weighing 64 kg, exercising twice a week, and without any reported upper extremity injuries) provided informed consent and willingly wore the exoskeleton and wearable IMU system. The individual proceeded to complete several repetitions of the over-head work task, following the guidelines of the Research Ethics Board at the University of Waterloo (ethics clearance: #43980).

Before the task-based experiments, the participant was briefed on the procedure and practiced the motions to ensure consistency. The wearable IMU system was calibrated using a standard pose (T-pose) to minimize sensor drift. All trials were conducted under consistent environmental conditions, with room temperature controlled at 19 °C.

The participant received instructions to lift a 2.267 kg kettlebell from a desk, upward until it forms around 90 ° elevation, and subsequently return it to the desk; verbal commands were provided to initiate these tasks (as indicated in Figure 1(a)). Furthermore, the participant engaged in training for an over-head work task (without load) involving raising the arm over-head and maintaining it in that position for an extended duration. These tasks were repeated 10 times with a 3 min rest period between repetitions.

The experimental setup, including sensor placement and exoskeleton alignment, was carefully checked before each trial to maintain measurement consistency. The 3-minute rest between repetitions was strictly followed to prevent muscular fatigue and ensure

uniform performance.

The motion data captured using the IMU system and onboard angle sensors of the BLDC motor were utilized as angle data in this study. The joint angles underwent a smoothing process through 30 Hz low-pass filtering to eliminate high-frequency noise, following a basic Fast Fourier Transform analysis. Subsequently, the angular velocity and acceleration were calculated numerically using the first and second-order numerical derivatives of the recorded joint angles. This processed data was then inputted into the human-exoskeleton model for the purpose of inverse dynamic simulation. The output of this simulation includes muscle torque generator excitation, activation signal, and joint torque exerted by the human participant. Additionally, the generated electricity within the BLDC motor was calculated using the identified model.

4 Results and discussion

4.1 Experimental identification. In Figure 3, the experimental data from the dynamometer and digital multimeter is presented. Figure 3(a) displays the passive torque of the BLDC motor, Equation (8), primarily resulting from the friction and inertia of the motor shaft, gearbox, and bearings. Figures 3(b) and 3(c) provide the angle and angular velocity, respectively, as measured by the dynamometer device. Although intended to operate at a constant speed, due to limitations in the range of motion, the dynamometer device experienced small delays in changing direction and attaining the desired speed. Figure 3(d) depicts the resistor voltage, recorded by the digital multimeter. Utilizing a 100 Ω resistor (with color coding of brown, black, brown, and golden; a power rating of 0.5 W; and an approxi-

mate maximum current of 71 mA), Figure 3(e) presents the calculated resistor current in relation to the filtered voltage (cleared high-frequency noise). For estimating the generated current, the constant consumed current for the on-board controller was subtracted from the total calculated resistor current. To achieve this, we determined the local mean current for each phase where the dynamometer maintained a constant angular velocity, then subtracted it from the total value, and provided the results in Figure 3(f). This procedure was performed for all investigated angular velocities.

Synchronization between the digital multimeter and the angular velocity of the dynamometer was achieved using convolutional techniques, which are mathematical operations that combine two signals by sliding one signal over the other and calculating the degree of overlap at each position. This process helps identify the optimal alignment between the signals, enabling accurate synchronization despite potential timing delays or inconsistencies in the measurements.

To further enhance measurement accuracy, we deliberately excluded the data at the very beginning and end of each trial. This decision was driven by the presence of initial transients—specifically, the inaccuracy introduced by the capacitor in the BLDC motor controller and the small initial or final jerk from the dynamometer, when starting, changing, or stopping the motion. Once the system reached a steady state in the measurements, we focused solely on the constant, stable portions of the data where reliability is maximized. This approach is substantiated by the consistent plateaus observed in the estimated generated current, as depicted in Figure 3(f).

Although perfect synchronization remains challenging due to the inherent differences in the data acquisition methods of the instruments, our strategy minimizes transient-induced errors. By ignoring the inconsistent initial and final segments of each measurement, our analysis is based on data that reflects the system's stable performance, thereby enhancing the overall accuracy. Feng et al [10] investigated energy regeneration from electromagnetic induction in lower-extremity robotic prostheses, where ensuring accurate signal alignment was crucial to isolating steady-state behavior. Similarly, Wu et al [22] addressed the challenges of capturing consistent energy harvesting data from a knee exoskeleton by focusing on the stable portion of the measurements and filtering out initial transient effects.

4.2 Relation of angular-velocity with generated-current. In this study, the relationship between angular velocity (expressed in $^{\circ}/s$) and the absolute generated current (measured in A), Equation (7), was investigated using MATLAB's polyfit function and illustrated in Figure 4(a). The data had angular velocity (ω) as the independent variable and generated electricity (I) as the dependent variable. Using the polyfit function, a Vandermonde matrix was created to solve the linear system. Centering and scaling techniques improved reliability, leading to a robust polynomial fit.

$$I = |p_3\omega^3 + p_2\omega^2 + p_1\omega| \quad (13)$$

where

- I the absolute generated current
- ω the angular velocity
- p_3 the cubic term of coefficients (with 95% confidence bounds) $4.967e-12$ ($-8.402e-13$, $1.077e-11$) (in $As^3/^{\circ}3$)
- p_2 the quadratic term of coefficients (with 95% confidence bounds) $-2.031e-09$ ($-3.674e-09$, $-3.886e-10$) (in $As^2/^{\circ}2$)
- p_1 the linear term of coefficients (with 95% confidence bounds) $2.951e-07$ ($1.66e-07$, $4.242e-07$) (in $As/^{\circ}$)

The obtained curve fit model (Equation (13)), a third-degree polynomial, reveals significant coefficients for each term, indicating a notable relationship between angular velocity and absolute generated current. The coefficient for the cubic term ($p_3 = 4.967e-12$) suggests a small but positive contribution to the generated electricity as angular velocity increases cubically. In contrast, the

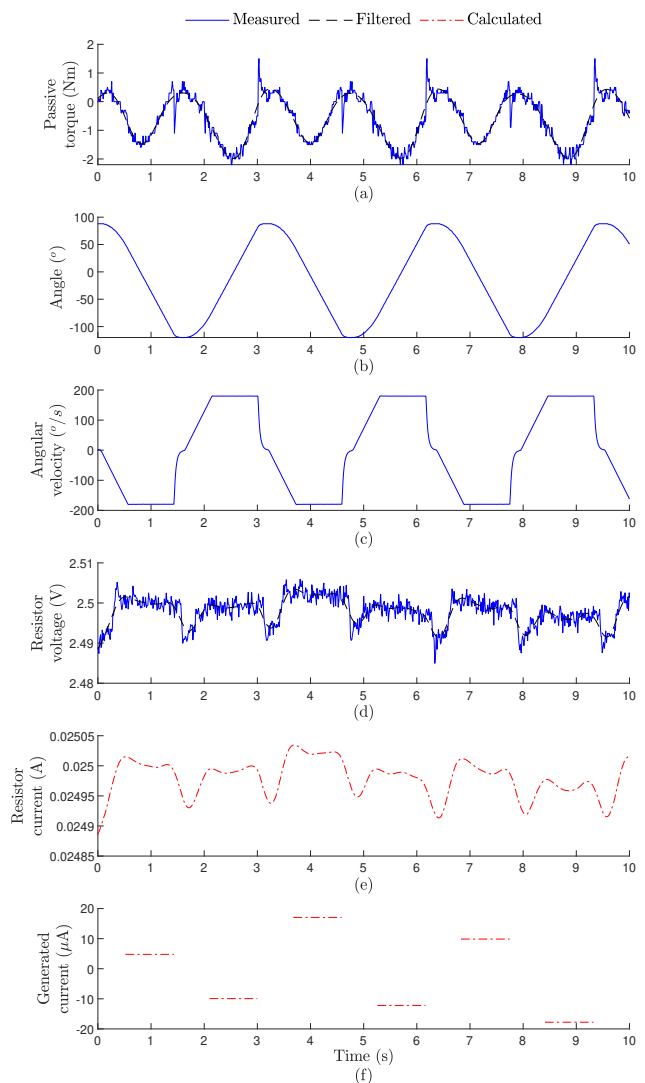


Fig. 3 A sample plot showing measured data from dynamometer and digital multimeter readings for the 180 $^{\circ}/s$ angular velocity passive test. The figure includes (a) passive torque, (b) angle, (c) angular velocity, (d) resistor voltage, (e) calculated resistor current, and (f) estimated generated current. Data representations: measured (blue solid line), filtered/processed (black dashed line), and calculated/estimated (red dashed-dot line).

quadratic term ($p_2 = -2.031e-09$) exhibits a larger negative coefficient, indicating a stronger effect where higher angular velocities result in reduced generated electricity. Lastly, the linear term ($p_1 = 2.951e-07$) shows a substantial positive influence on generated current with increasing angular velocity. These coefficients, along with their confidence bounds, illustrate a complex relationship where the motor tends to generate more current as it rotates at higher velocities, with the cubic and linear terms being the primary contributors to this trend. This conclusion aligns with expectations in electro-mechanical systems, where higher speeds generally result in increased electrical output due to the motor's enhanced performance and efficiency.

4.3 Relation of angular-velocity with passive torque. In Figure 4(b), the relationship between passive torque and angular velocity (Equation (8)) as well as efficiency is depicted. The same polyfit technique was employed to establish the linear relation. The observed slope of 0.001337 N m in the passive torque-angular ve-

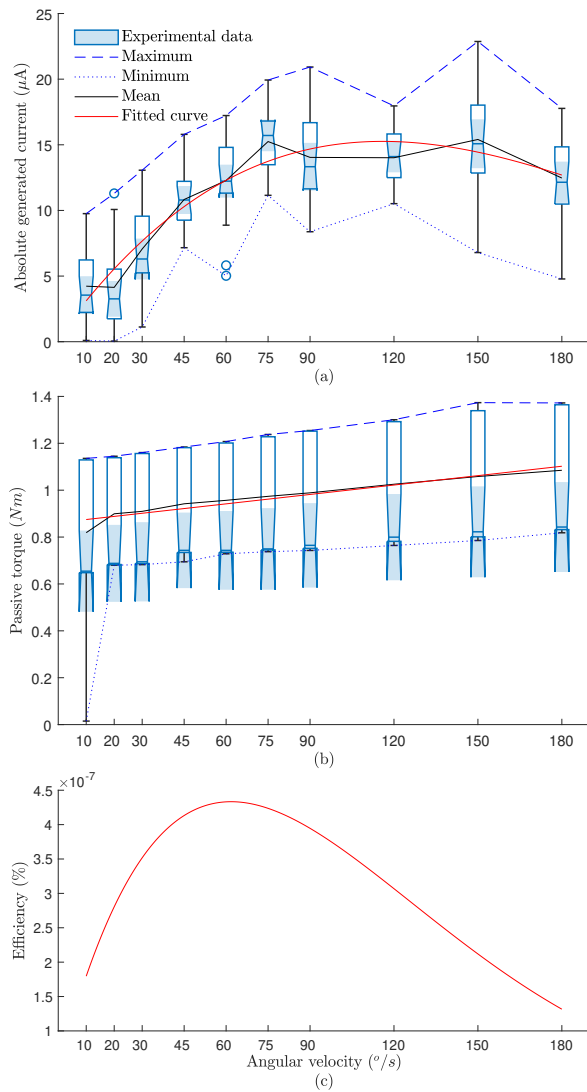


Fig. 4 Statistical relationship between angular velocity and (a) absolute generated current, (b) passive torque, and (c) efficiency. Data representation includes: the mean measurement (black solid line), maximum (blue dashed line), minimum (black dotted line), fitted curve (red solid line), the upper (0.75) quartile and lower (0.25) quartile (blue solid boxes with notches), and the interquartile range (blue shaded region).

locity relationship, Equation (8), can be primarily attributed to the frictional forces within the BLDC motor. Friction is a critical factor influencing the overall performance and efficiency of mechanical systems, particularly in rotational machinery like motors. The presence of friction introduces resistance to motion, resulting in the generation of torque opposing the motor's rotation. This phenomenon is commonly known as "passive torque".

In Figure 4(c), the efficiency of the BLDC motor is computed using Equation (11) with the fitted curves for torque and current. It is evident that this efficiency is quite low (ideal operating speed is about $60^{\circ}/s$), indicating that the electrical energy regeneration may not be beneficial under normal operating conditions. Nevertheless, in remote areas or military applications, even this modest efficiency becomes worthy of consideration [4]. Note that this efficiency is specific to the model of motor analyzed in this work and would vary for different motor types.

Several factors contribute to the observed low efficiency of the BLDC motor in regenerative mode. One major source of loss is

mechanical friction in the motor's bearings and gear train, which dissipates energy as heat. Additionally, resistive losses in the windings increase energy dissipation, particularly at low speeds where current draw is relatively high compared to power output. Another key factor is conversion inefficiencies within the motor controller, which can result in further energy loss due to rectification, pulse-width modulation switching, and electrical resistance in the power electronics. These losses are typical in low-speed regenerative systems, as seen in previous studies on energy recovery in small-scale actuators [40–42].

Furthermore, the specific design of the wearable exoskeleton influences efficiency. Unlike industrial motors optimized for continuous operation at high efficiency, exoskeleton motors prioritize back-drivability and user comfort. This means that design trade-offs, such as lower gear ratios and reduced mechanical impedance [8], inherently limit the potential for efficient energy conversion. Additionally, since the measured voltages are extremely low, small variations in resistance, contact losses, and environmental conditions can disproportionately affect the efficiency calculation. While further optimizations—such as improving controller algorithms or using ultra-low-friction components—could enhance efficiency, the current findings provide a realistic assessment of energy recovery potential in wearable robotics applications.

4.4 Task-based simulation results. Figure 5 illustrates the experimental joint angle of the shoulder (e.g., flexion/extension, abduction/adduction, medial/lateral rotation), the elbow (flexion/extension), and forearm (supination/pronation) during object manipulation (left) and over-head work tasks (right). We use mean values from repeated task executions.

Using Equation (12), the computed energy expenditure for each task is depicted in Figure 6. The motion profiles captured in Figure 5 are regarded as standard speed tasks, while speed variations of 50, 150, and 200% of their normal values are applied to symbolize slow, fast, and very fast motion, respectively. The calculated energy ranges from $3.04e-5$ to $1.02e-3$ J for tasks corresponding to slow to very fast motion scenarios.

The generated electrical energy by the BLDC motor operating in passive mode while being driven by human motion has been measured for two tasks. While the amount of energy generated in this manner is relatively low, especially on an individual basis, it becomes significant when considering repeated and fast motions over time. For instance, soldiers wearing exoskeletons during missions can harness this generated energy to charge essential electrical equipment, such as communication devices or sensors, thereby enhancing their operational capabilities in remote locations [43,44].

Note that the passive torque exhibited by the motor was significantly lower compared to the total torque exerted by the human shoulder, posing no issues regarding the required passive torque. If the passive torque exhibited by the motor had been very high and not back-drivable, it would have required the user to carry a heavy object or provide excessive torque to move the motor. This phenomenon can be attributed to either the low gear ratio of the motor or its direct drive feature, both of which are essential for its utilization in assistive exoskeleton applications [14]. While acknowledging the importance of maintaining a low passive torque, it is important to highlight that a higher gear ratio could enhance the motor's energy generation capabilities, as it would result in a higher rotational speed of the rotor.

The ability to utilize the electrical energy generated by BLDC motors in passive mode showcases the potential for sustainable power solutions in demanding environments. Through efficient energy conversion and utilization technologies, even small amounts of generated energy from human motion can be harnessed to power critical equipment, reducing reliance on traditional power sources and increasing operational autonomy.

4.5 Data analysis and comparison. Analysis of how angular velocity influences the generated current and passive torque is as

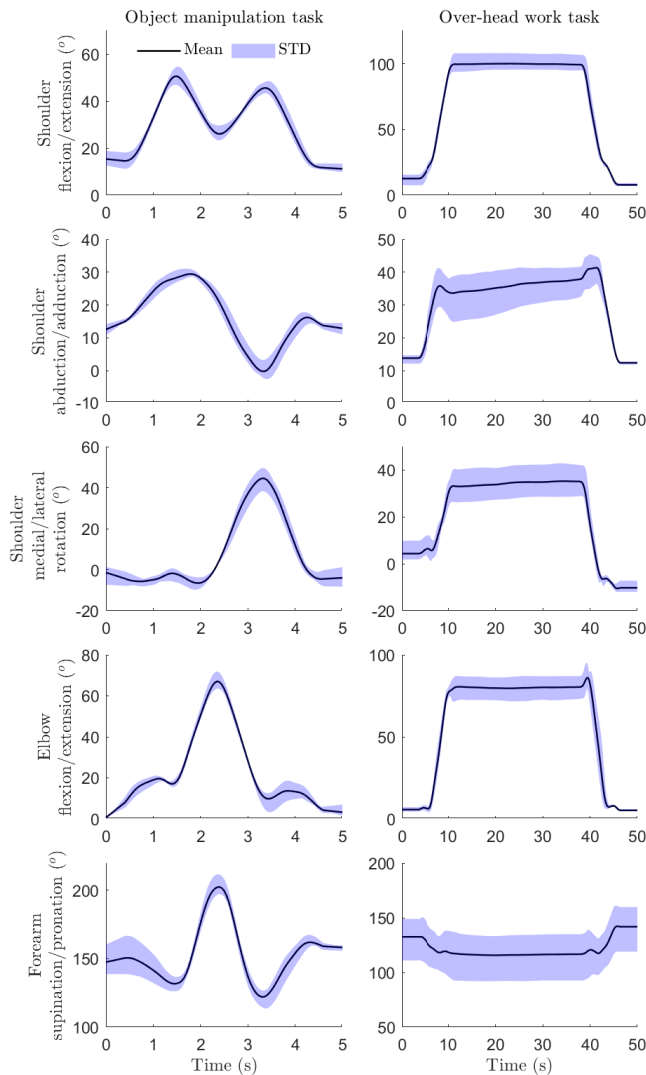


Fig. 5 The relative angular rotation of shoulder movements encompassing flexion/extension, abduction/adduction, and medial/lateral rotation, along with elbow flexion/extension and forearm supination/pronation, during object manipulation (left) and over-head work tasks (right). Data representations: mean measurement (black solid line) and standard deviation (blue shaded bands).

follows. As shown in Figure 4(a), the generated current exhibits a nonlinear increase with angular velocity, as captured by our third-degree polynomial fit (Equation (13)). The positive linear and cubic terms indicate that higher angular velocities generally drive an increase in current generation, while the negative quadratic term suggests that efficiency diminishes beyond a certain speed due to increased frictional losses. In parallel, Figure 4(b) illustrates that the passive torque increases linearly with angular velocity (with an observed slope of $0.001337 \text{ Nms/}^\circ$), a behavior consistent with frictional effects within the BLDC motor. This combined evaluation underscores that while increasing angular velocity boosts energy generation, there exists an optimal operating range where the effects of friction are minimized. Figure 4(c) illustrates how efficiency peaks around $60^\circ/\text{s}$, forming a single-lobed curve. This indicates a “sweet spot” where the trade-off between increased current generation (benefiting from faster rotation) and rising frictional losses is most favorable. Beyond this point, efficiency declines as frictional effects dominate any additional current gains.

Figure 4(a) shows that while the absolute generated current gen-

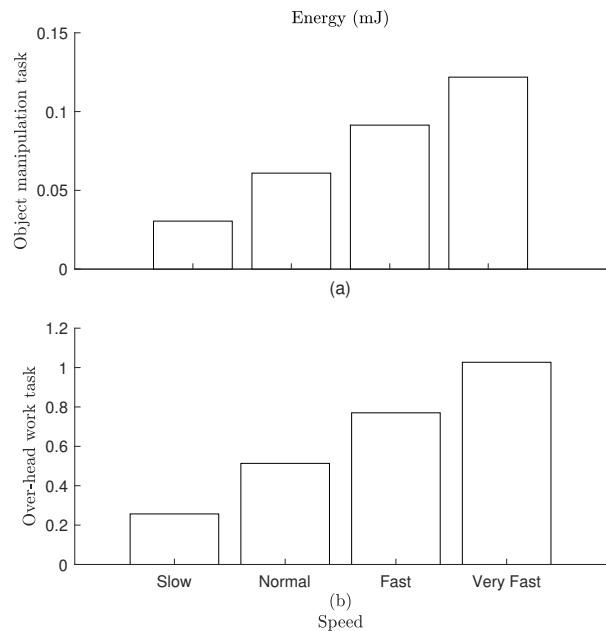


Fig. 6 The energy generated for a single repetition of object manipulation task (a) and over-head work task (b) at varying speeds: slow, normal, fast, and very fast.

erally increases with speed, variability across repeated trials becomes more pronounced at higher angular velocities. The widening gap between maximum and minimum measured currents suggests that factors such as slight variations in friction or initial conditions affect consistency. Despite this, the polynomial fit effectively captures the overall trend, highlighting that further increases in speed offer diminishing returns in current generation. In Figure 4(b), the vertical bars representing torque variability remain relatively narrow, indicating consistent frictional behavior across tests and reinforcing the reliability of the passive torque measurements. Lastly, Figure 4(c)’s efficiency curve underscores that operating beyond the $60^\circ/\text{s}$ threshold yields declining efficiency, emphasizing the importance of balancing speed and frictional losses to maximize performance.

Our task-based simulation results reveal clear performance differences under varied experimental conditions. As depicted in Figure 6, the energy generated per repetition varies markedly with speed—ranging from approximately $30.4 \mu\text{J}$ for slower movements up to 1.02 mJ for very fast motions. Additionally, the comparison between object manipulation and over-head work tasks (Figure 5) demonstrates that the nature and dynamics of the movement significantly affect energy regeneration. Specifically, the over-head work tasks, which involve more sustained and higher-range motions, consistently produce higher energy outputs compared to the object manipulation tasks. This analysis not only highlights the importance of motion characteristics on energy harvesting but also emphasizes the potential for optimizing operational parameters (such as gear ratio, transmission settings, motor drive parameters) to maximize the system’s overall efficiency.”

4.6 Limitations of this work. Limitations in the current study primarily revolve around the experimental setup and the scope of analysis. Firstly, the study’s focus on a specific type of upper-limb exoskeleton and a particular BLDC motor model limits the generalizability of the findings to other similar systems. Variations in exoskeleton designs and motor specifications could yield different results regarding energy generation and system performance.

Another limitation is the controlled laboratory environment in which the experiments were conducted. Real-world scenarios may

introduce additional variables and challenges that were not accounted for in the study. Factors such as different user behaviors and extended periods of use could impact the energy regeneration capabilities and overall system performance of wearable robots.

Furthermore, the study mainly focuses on the technical aspects of energy generation and system dynamics, without delving deeply into user experience, comfort, and long-term usability. Future studies could incorporate user feedback, ergonomic evaluations, and extended wear trials to assess the practicality and acceptability of regenerative wearable robots in real-world settings.

Another limitation of the current study is the inherent challenge in measuring extremely low voltage levels generated during the regenerative process. Although our measurement device is designed to resolve these low voltages with high precision, even small inaccuracies may occur. However, our identification of the current–angular velocity relationship using the third-order polynomial model (Equation (13)) has been repeated over multiple experimental runs, and the reproducibility of the results confirms that any error remains below 10%. Such minor errors do not significantly affect the simulation-based estimation.

Moreover, while alternative models could be considered, our tests indicate that the third-order polynomial model reliably captures the relationship between angular velocity and generated current within the operational range. Future work will investigate additional modeling approaches and further optimize the measurement setup. This would help ensure greater accuracy when dealing with very low voltage measurements, though the present inaccuracy is not critical to the overall findings of the study.

4.7 Challenges and future work. The development of regenerative wearable robots poses numerous challenges that necessitate careful consideration for their successful and safe integration. These challenges exhibit parallels with those encountered in the development of hybrid vehicles, underlining the intricate nature of merging regenerative technologies into wearable robotic systems.

One main challenge lies in achieving seamless transitions from assistive to regenerative modes within the robot controller. This necessitates the implementation of sophisticated algorithms and control systems capable of ensuring smooth and efficient operation while accommodating the user's intentions and movements [16,27]. For instance, utilizing surface electromyography signals can facilitate the identification of user intent [36], enabling the robot to switch modes when it discerns that assistance is no longer required based on the user's intent, thereby enabling regenerative capabilities.

The shift in human movement patterns when focusing on regenerative capabilities presents challenges in terms of user adaptation and comfort. Designing wearable robots that complement and enhance natural human movements, rather than impeding them, is essential. Consequently, studying the central nervous system during the user's adaptation to the new regenerative system is important [30,45,46].

Safety considerations are important, particularly during tasks involving heavy objects, as the regenerative capabilities may not be safe during sudden downward force applications. Ensuring that the system can manage such scenarios without compromising safety is critical for real-world applications [47,48].

Additionally, storing the electrical energy generated through regeneration poses technical hurdles due to the low energy densities of current storage technologies. Exploring efficient and compact energy storage solutions is imperative to render regenerative wearable robots practical and functional for extended periods. Furthermore, isolating power from the BLDC on-board controller is essential to avert potential hazards and ensure the overall safety of the system.

In terms of future work, several areas warrant further exploration. Enhancing energy storage solutions to accommodate higher energy densities and longer usage durations is crucial for advancing regenerative wearable robots. Additionally, developing intelligent control algorithms that can dynamically adjust assistance and regeneration levels based on user intent and task requirements would significantly improve system efficiency and user experience.

5 Conclusions

In conclusion, our study demonstrates that implementing backdrivable actuators with energy regeneration capabilities can slightly enhance the efficiency and prolong the battery life of robotic upper-limb exoskeletons. Through converting a portion of dissipated energy during negative mechanical work into electrical energy, we achieved a small amount of energy regeneration. By conducting task-based experiments during daily-living activities of the exoskeleton, we were able to assess the effectiveness of this energy regeneration approach.

The challenges in developing regenerative wearable robots include seamlessly transitioning between assistive and regenerative modes, adapting to changes in human movement patterns, ensuring user safety during heavy tasks, and addressing energy storage limitations. New algorithms and control systems are needed to manage mode transitions and user intent, while studying the central nervous system aids in user adaptation. Exploring compact energy storage solutions and ensuring system safety are needed for practical and functional regenerative wearable robots.

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Institutional review board statement

The study was conducted according to the guidelines of the Declaration of Helsinki, and approved by the Office of Research Ethics of the University of Waterloo #43980.

Abbreviations

BLDC brushless direct current.

DoF degree of freedom.

EMF electromotive force.

IMU inertial measurement unit.

MSK musculoskeletal.

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